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Development of intelligent design platform for robotic drilling system based on CBR

Ying Xu*, Yue Zhang, Zhanghuan Xu, Xianghu Zeng, Jianqiang Chen

School of Aircraft Engineering, Nanchang Hangkong University, Nanchang, China

32006@nchu.edu.cn

Abstract. In order to improve the design efficiency of the robotic drilling system, the case-based reasoning, CBR design technology is used in the development of the robotic drilling system design platform. The framework of the design platform of the robot hole-making system is constructed, and the key technologies for the construction of the intelligent design platform are studied, including the graphics similarity judgment method, parts coding rules, part parameterization rules, software and platform interface technology, drilling end effector geometric modeling, etc. Combined with the end effector design example, the effectiveness of this method is verified.

1. Introduction

In the rapidly developing environment of networking, informationization, digitization, automation and intelligence, aircraft design, manufacturing and training technology have also developed rapidly [1,2,3]. Although three-dimensional models have been widely used in many aspects of aircraft design and manufacturing, the meaningful smart design of which has not yet been realized [4]. Due to the special shape and structure of aircraft products, the complexity of the assembly process and the uncertainty of the hole-making process, the designers of drilling end effectors often involve time-consuming, repetitive labor, low efficiency, and difficult to meet the design qualities for the high precision requirements of the hole position [5]. In order to improve this situation, it is necessary to use intelligent design methods to utilize the inheritance and reproduction of design knowledge to support the improvement of the aircraft assembly automation process [6,7,8,9].

Aiming at the complexity and low efficiency of the design process of robotic end effector, a design method of hole-making end effector based on case-based reasoning, namely CBR, is proposed. The system design framework is constructed based on the overall functional division of the system. The key technologies for the construction of the drilling end effector design platform are studied, including the graphics similarity judgment method, the part coding rules, the part parameterization rules, modular reasoning rules and drilling end effector geometric modeling, etc. A parameterized model library of the design platform is created, combined with end effector design examples, to verify the effectiveness of this method.

2. CBR methods and steps to solve problems

Case reasoning refers to the past cases stored in the case database to correct the original case to solve the problem. The steps to solve the problem are as follows:

(1) Retrieve: Similar cases retrieved and extracted from the case database for the problem.

(2) Reuse: Information contained in the selected similar cases is used to solve the corresponding problems.

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(3) Revise: The case is revised appropriately to get a new case.

(4) Retain: In order to provide reference for subsequent problem solving, the new instance is stored in the instance library.

Different from rule reasoning, in case reasoning, problem reconciliation is stored as an instance, and the experience of solving the problem is only implicit in the instance. Figure 1 shows the design algorithm for the robotic drilling system based on CBR.

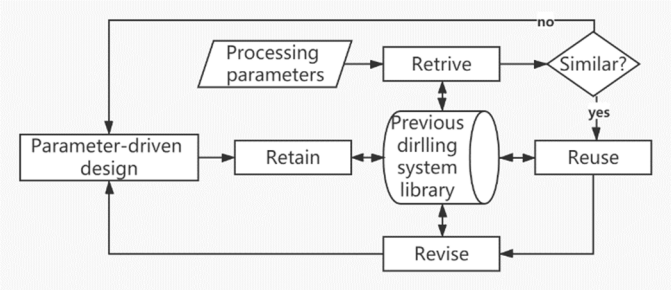


Figure 1. Robotic drilling system design algorithm based on CBR

3. The architecture of the design platform

The design platform of the robotic drilling system is a man-machine interactive system. The application of product platform technologies applied to robotic drilling system include using VB6.0, building the component structure model through SolidWorks, and choosing Access to create the database. The structure of the design platform of the robotic drilling system is shown in Figure 2.

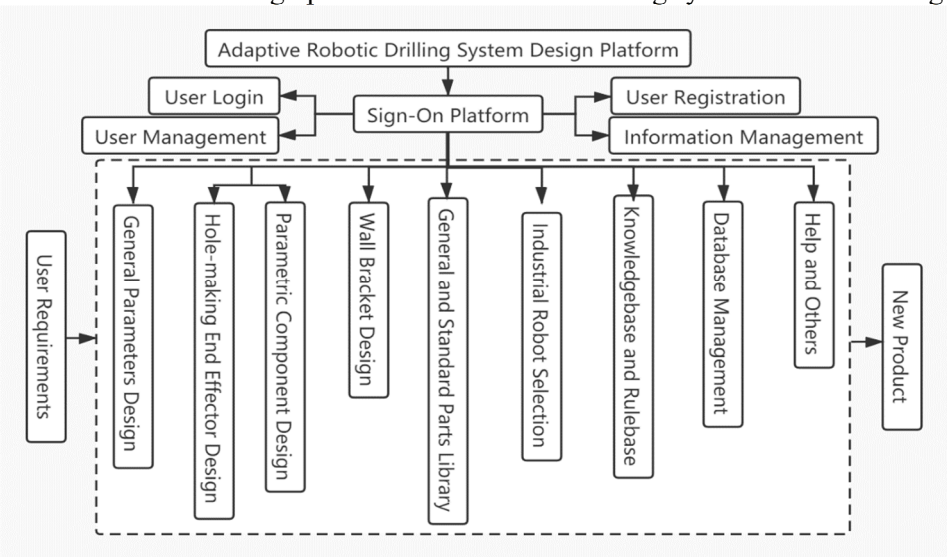


Figure 2. Robotic drilling system architecture

4. The main functions of the platform

The main functions of the platform include human-computer interaction, data storage, design scheme fitting, design result display output, etc. The functions are as follows:

(1) General parameter design

The General design parameter interface is divided into four areas, namely design parameter input area, case information area, parameter table area, and parameter management area. The design requirements input query whether there are cases that meet the design requirements. With existing case, you can learn from the design and update the case database, otherwise start the new design.

(2) Drilling end effector design

The design of the drilling end effector is the core of the hole-making system design platform, and its main components are detection unit, compression unit, drilling unit, etc. [10,11].

Detection unit: the detection unit mainly includes reference detection and normal detection. The reference detection is composed of a high-precision industrial camera and a special light source, and the normal measurement is composed of four symmetrically distributed laser ranging sensors.

Compression unit: The compression unit is a self-made unit, shown in Figure 3.

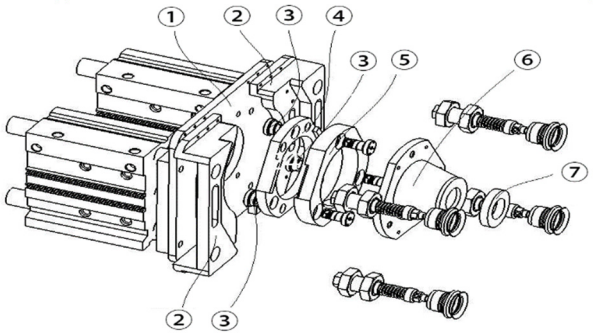


Figure 3. Compressing Unit. ①Pressure-plate ②Sucker Mount ③Joint Sleeve ④Pressure Support ⑤ Flange Connection ⑥Compressing Nose ⑦Spring Washer

Driller unit: It is composed of a feed unit and an electric spindle, including a base plate, a servo motor, a transmission mechanism, a coupling, a ball screw, a linear slide, and a motor spindle.

(5) Robot selection

Mainly include: ①Management of industrial robot products; ②Display of selection results; ③ Installation requirements; ④Saving of selection results.

5. The key technologies used in platform development

5.1. Development Process

The waterfall model is adopted for development, as shown in Figure 4.

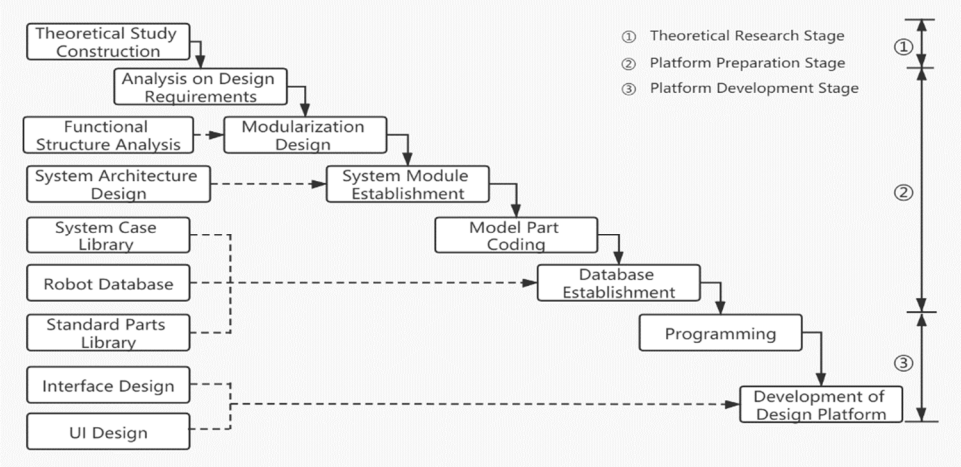


Figure 4. The Development Process of the Design Platform

5.2. Key Points of Platform Development

(1) Interface between platform and commercial software

The software interface of the design platform of the robotic drilling system includes two types: the interface of the platform and the database, the interface of the platform and the modelling software.

(2) Parts encoding

The module encoding principles adopt a flexible encoding system, shown in Figure 5.

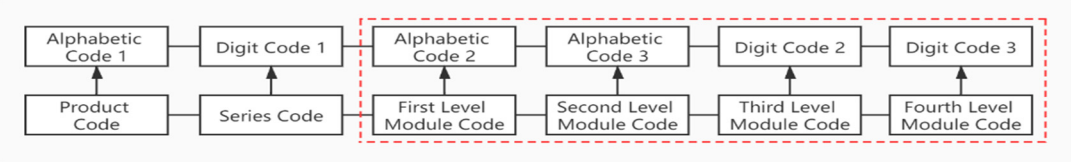


Figure 5. Encoding structure, rules: a. Structural product code, consists of two letters, such as hole-making end effector (MZ); b. Structure series code, composed of two numbers, 00,01,02...; c. First Level module code, A, B, C...; d. Second level module code: a, b, c...; e. Three level module code: 0,1,2...; f. Four level module code, 00, 01, 02...; g. The code in the dashed box can be omitted according to needs.

(3) Parametric design of parts

Designing components on the design platform is a process of parameter-driven model changes, so the design of the parametric model library is the foundation of the platform. This article selects the self-programming method using interface technology to control the operation of SolidWorks.

(4) Judgment of similarity

Retrieving the processing instance database, using the three-dimensional model of the processing object as the processing feature recognition, the results bring out the instances with similar processing features from the database. Using the processing data to determine the processing features judge the similarity of all the extracted processing features.

(5) Establish a platform database

Each of the system includes the manufacture information of the new manufacture case, and the manufacture data are used when processing the manufacture data via the highest comprehensive similarity with the same manufacture features retrieved from the database. The manufacture data used by CBR judging the similarity are 12 parameters such as the tool used in processing, the number of spindle revolutions, and the feed speed, etc. [12]. The database structure is shown in Figure 6. The part model database has a large amount of data, such as standard parts library and self-made parts library.

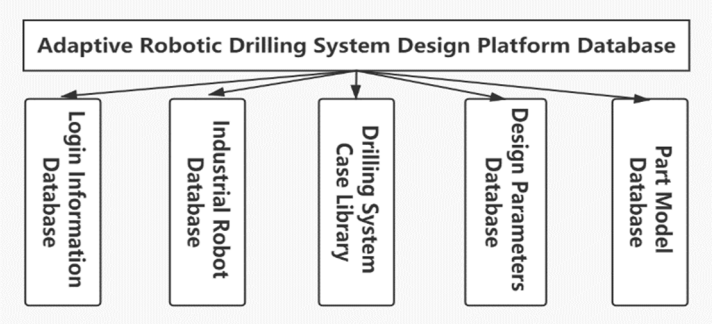


Figure 6. Design Platform Database Structure

6. Example design based on the design platform

According to the example requirements, a robotic drilling system demo is designed by the platform. In the general parameter design interface, shown in Figure 7, there are two ways to enter the design. One is to enter the general requirements and parameters, and enter the design program as required, which is the main function. The second is the design case query, which is an auxiliary function. After the designer enters the design parameters, click "Write" to write the parameters in the parameter table, and then click "Save" to save and enter the design.

If you click the "Query" button, the background program based on CBR operates and the output query results will be displayed, whether there are identical or similar cases, and provide an example reference for the design. You can click the "Start" button to enter the main interface of the drilling end effector design.

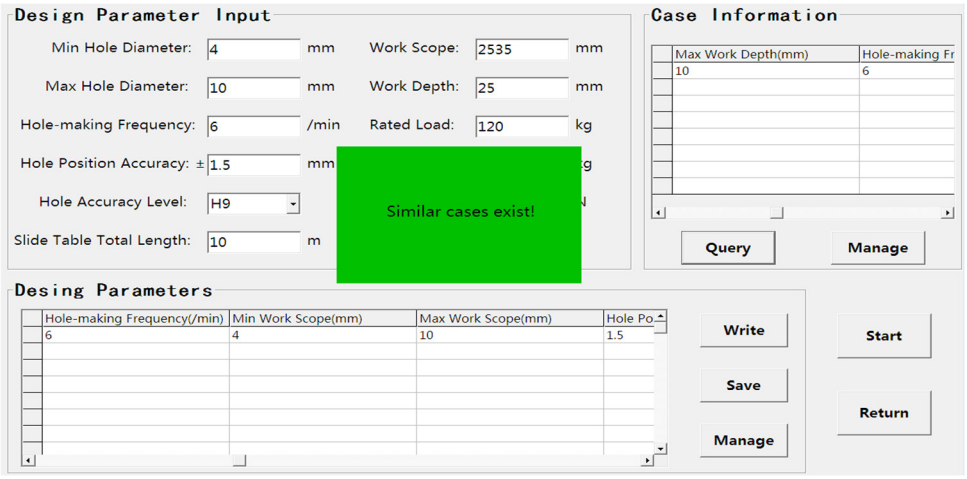


Figure 7. General parameter design interface

As shown in Figure 8, you can click the "driller design" button entering the driller design interface. The interface design includes standard parts selection, self-made parts selection, tool selection, installation dimensions, etc. After completing the design of the driller, click the "return" button, and then click the "compressing unit design" button to enter the compressing unit design interface.

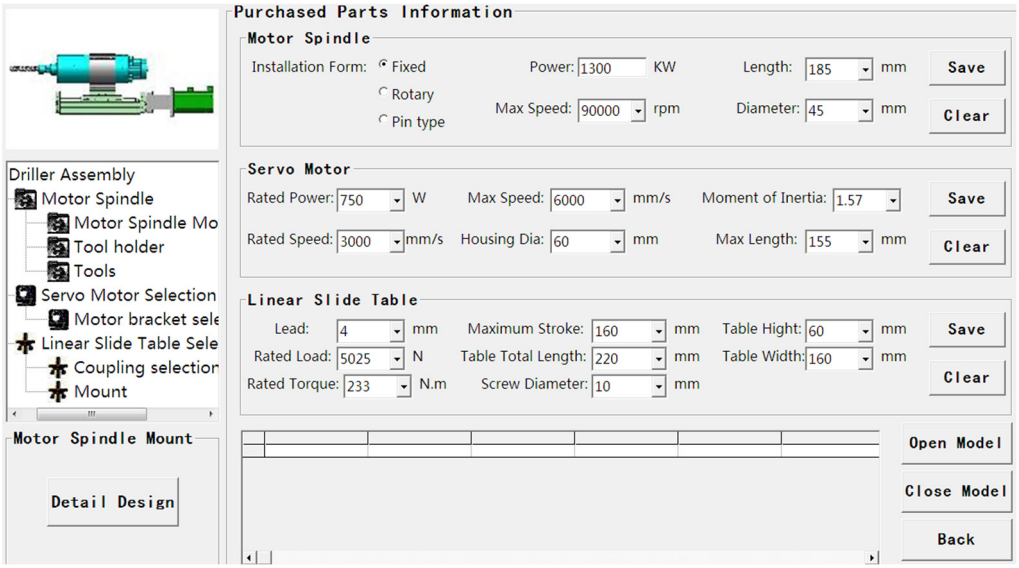


Figure 8. Driller Design Interface

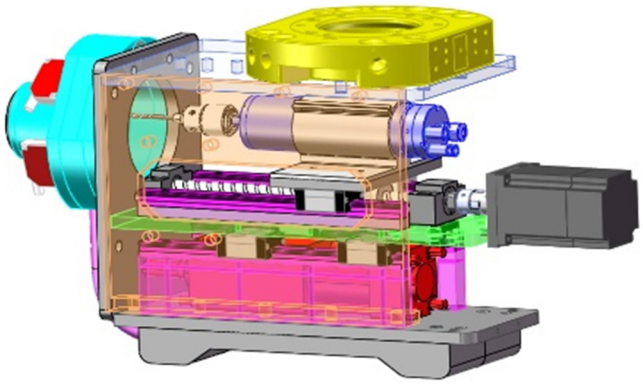


Figure 9. Assembly renderings of the Designed Drilling End Effector Demo

The bottom plate and top plate are designed in the pressure support unit design interface, and the parameters are input to implement the parameterized drive design. After saving the data, the model library can be opened for model browsing and selection. By analogy, you can design the detection unit and the chip suction and removal unit, and then select the industrial robot and connection mode of the end effector and the robot to complete the design. The finished demo is shown in Figure 9.

7. Conclusion

Based on the application characteristics of case-based reasoning (CBR) in intelligent design, a case study of the design of a robot hole making system based on CBR is studied. With the perspective of platform operating results, the utilization of the end effector geometric model and the case-based reasoning method base on its corresponding knowledge of product design improves the design concept of aerospace products and enhances the consistency of design and manufacturing data in the aircraft assembly process,improving design efficiency.

In future research, the author will further expand and improve the geometric library, rule library, and parameter case library, add performance analysis functions to the design platform, and lay the foundation for digital aircraft design and manufacturing.

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References

[1] Son, H., Choi, H.J., Park, H.W. (2010) Design and dynamic analysis of an arch-type desktop reconfigurable machine. Int. J. Mach. Tools Manuf., 50: 575-584.
[2] Shpitalni, M., Koren, Y. (2010) Design of reconfigurable manufacturing systems. J. Manuf. Syst., 29: 130-141.
[3] Xu, Y. (2019) A comparative and explore on the training mode of aviation talents between China and America. Sci. Innov., 7: 70-75.
[4] Xu, Y., Wan, Q., Raymond M.T., Lu G., Zeng X. (2019) Design of solar powered and fatigue durable quadcopter. Int. J. Mech. Eng. Appl., 7: 58-65.
[5] Olsson, T., Haage, M., Kihlman, H., Johansson, R., Nilsson, K., Robertsson, A., Björkman, M., Isaksson, R., Ossbahr, G., Brogårdh, T. (2010) Cost-efficient drilling using industrial robots with high-bandwidth force feedback. Robot. Comput. Integr. Manuf., 26: 24-38.
[6] Chen, J. (2017) Adaptive robot drilling system design platform construction[dissertation]. https://kns.cnki.net/kcms/detail/detail.aspx?dbcode=CMFD&dbname=CMFD201801&filename=1017711428.nh&v=woFh%25mmd2F1AEMal8Tpe0RJB0%25mmd2FxfGbqoDzRIF9qIlzYpRJdVZFF3pA6Y%25mmd2FIurCpHKNPslC.
[7] Xu, Y., Wan, Q., Lu, G., Zhang, B. (2018) Dynamic characteristics of the end-effector of a drilling robot for aviation. Int. J. Mater. Sci. Appl., 7: 192-198.
[8] Aziz, M.H., Ayub, M.A., Jaafar, R. (2012) Real-time Algorithm for detection of breakthrough bone drilling. Procedia Eng. 41: 352-359.
[9] Gallardo, J., Lesso, R., Rico, J.M., Alici, G. (2011) The kinematics of modular spatial hyper-redundant manipulators formed from RPS-type limbs. Rob. Auton. Syst., 59: 12-21.
[10] Zhang, L., Wang, X. (2012) A novel algorithm of normal attitude regulation for the designed end-effector of a flexible drilling robot. J. Southeast Univ., 1: 29-34.
[11] Yang, M., Gui, L., Hu, Y., Ding, G., Song, C. (2018) Dynamic analysis and vibration testing of CFRP drive-line system used in heavy-duty machine tool. Results Phys., 8: 1110-1118.
[12] Mi, J., Zhen, Z., Zhu, Y. (2018) Research on dynamic characteristics of machine tools based on high acceleration processing. J. Mech. Strength., 40: 987-991.